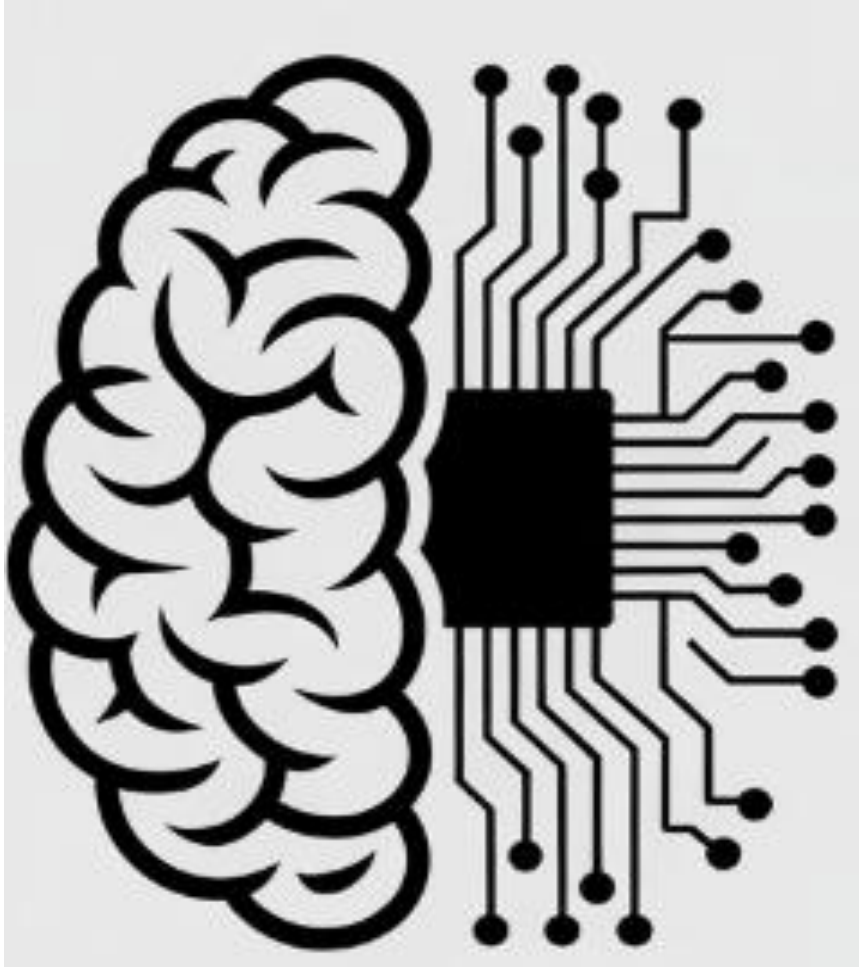




Alpha-Beta Pruning

Dr. Sobia Tariq Javed

Alpha-Beta Pruning

- Eliminating a branch without consideration is called pruning
- A way to improve the performance of Minimax procedure

- **Basic Idea:**

“If you have an idea which is surely bad, don’t take the time to see how awful it is” – Pat Winston

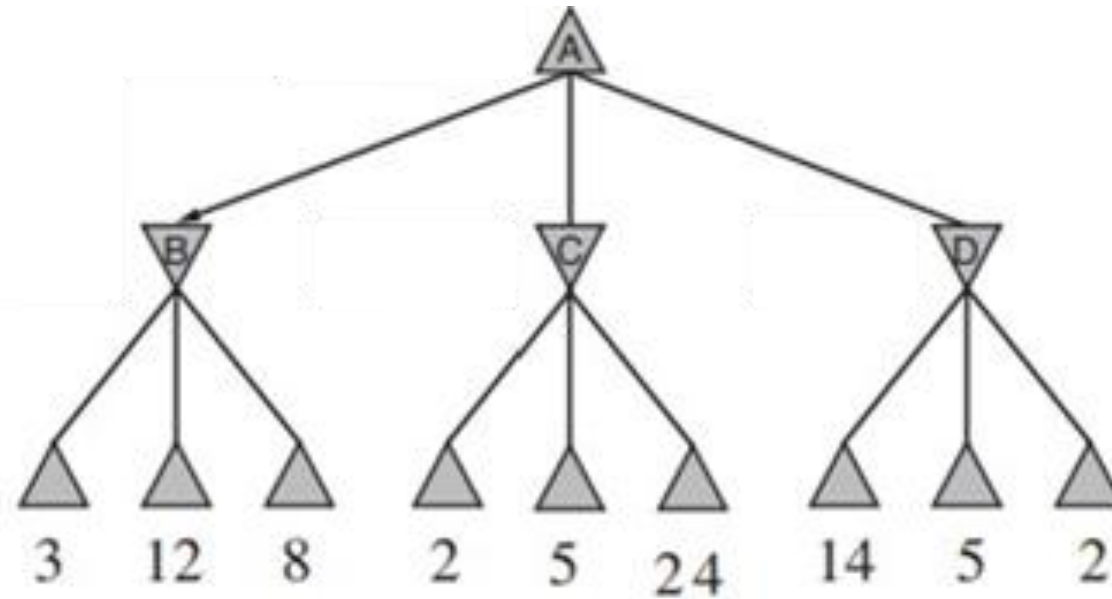
Alpha-Beta Pruning

- Proceed in a depth-first fashion in the n-ply look-ahead search tree.
- Find the score for the top of this tree.
- During the search, creates two values alpha and beta
- α -value associated with MAX can never decrease
 - MAX's eventually score is at least as good as the current α -value
- β -value associated with MIN can never increase
 - MIN's eventually score is at least as good as the current β -value
- α -prune: β -value $\leq \alpha$ -value of a MAX ancestor
- β -prune: Any MAX node having β -value $\leq \alpha$ -value of any MIN ancestor

Alpha Beta Pruning

Max

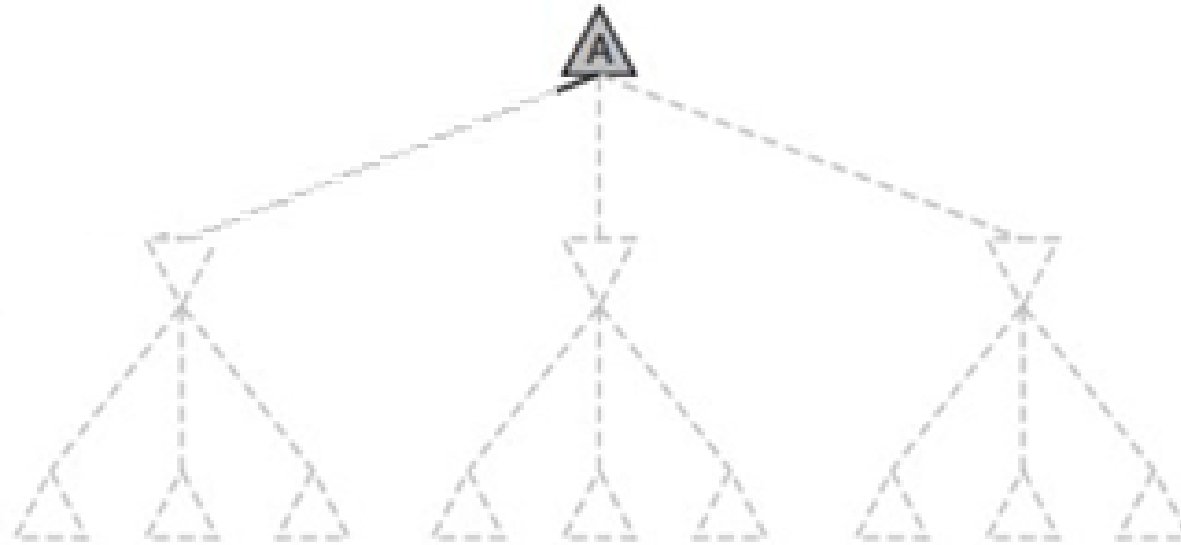
Min



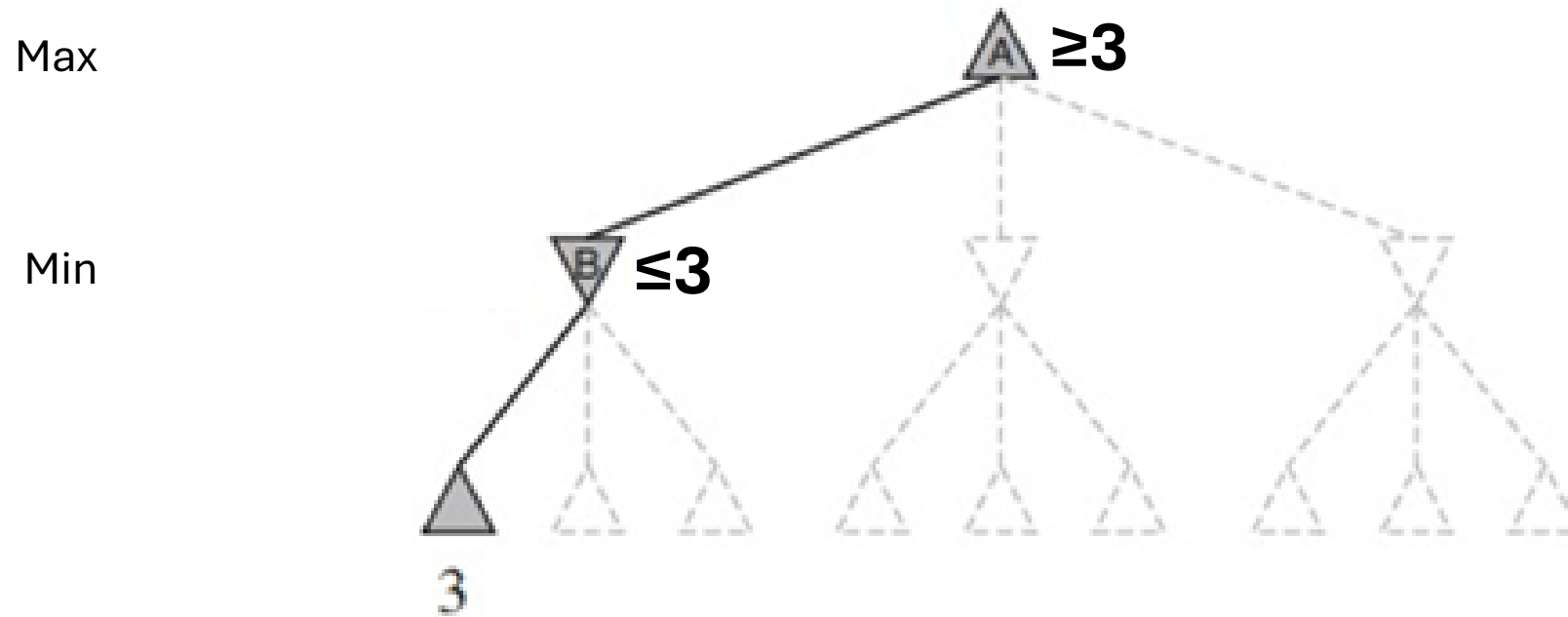
Alpha Beta Pruning

Max

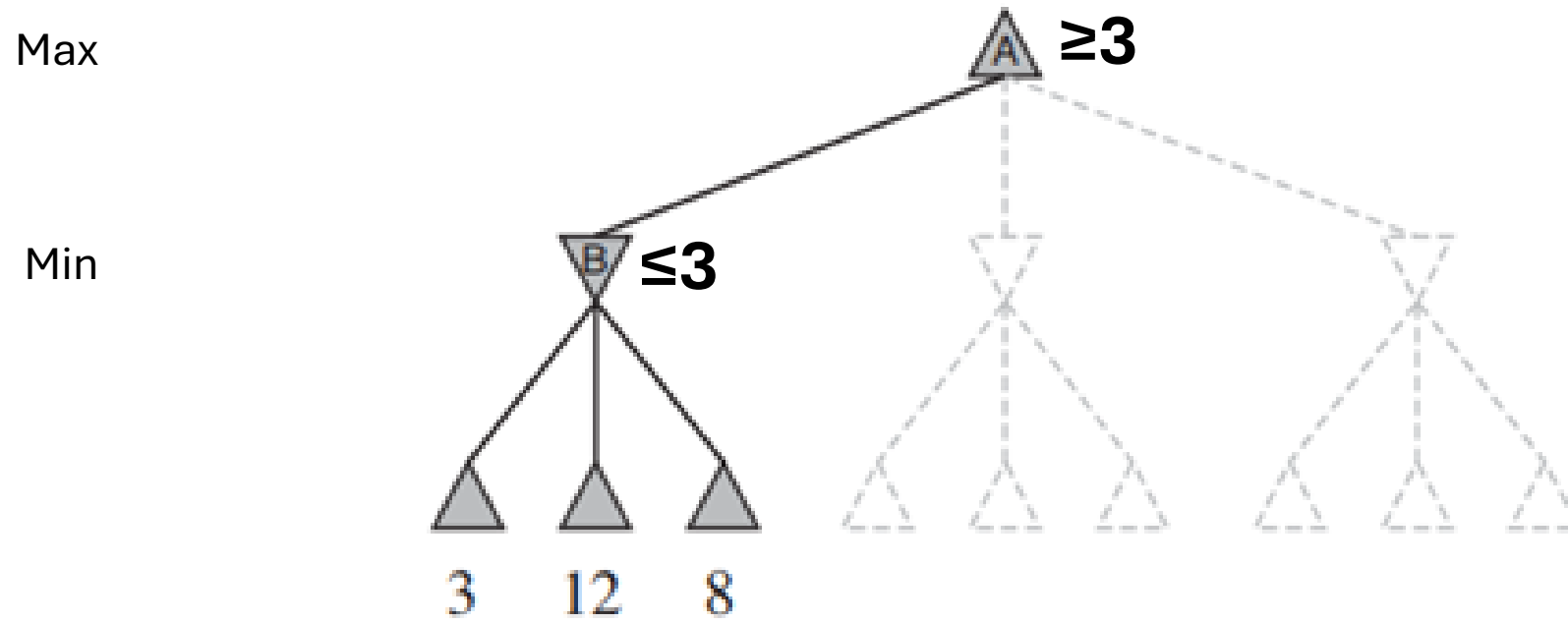
Min



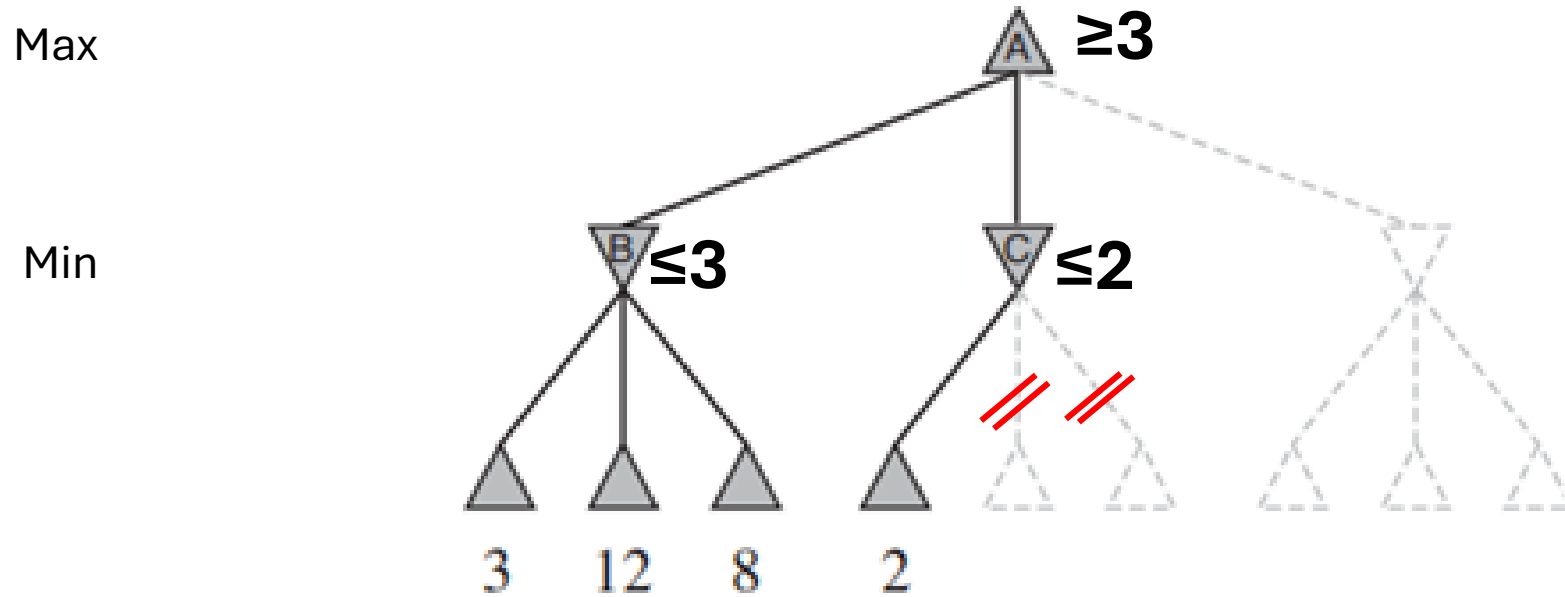
Alpha Beta Pruning



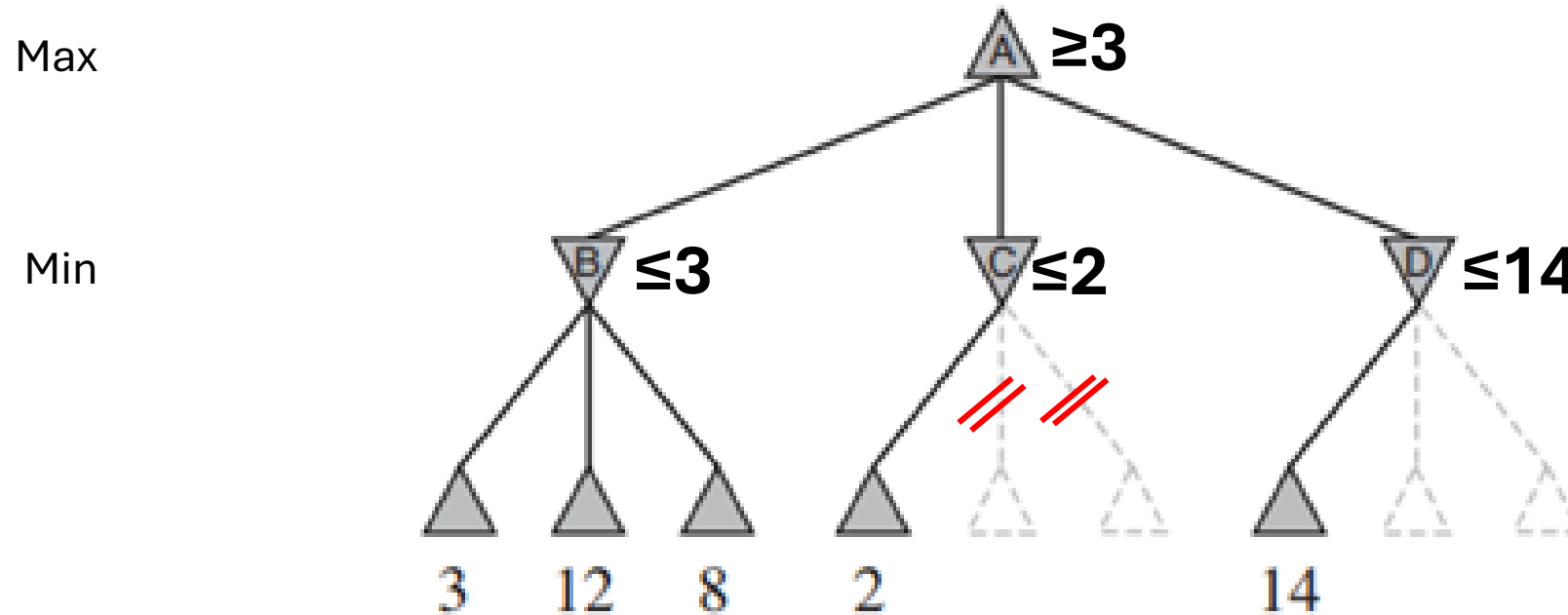
Alpha Beta Pruning



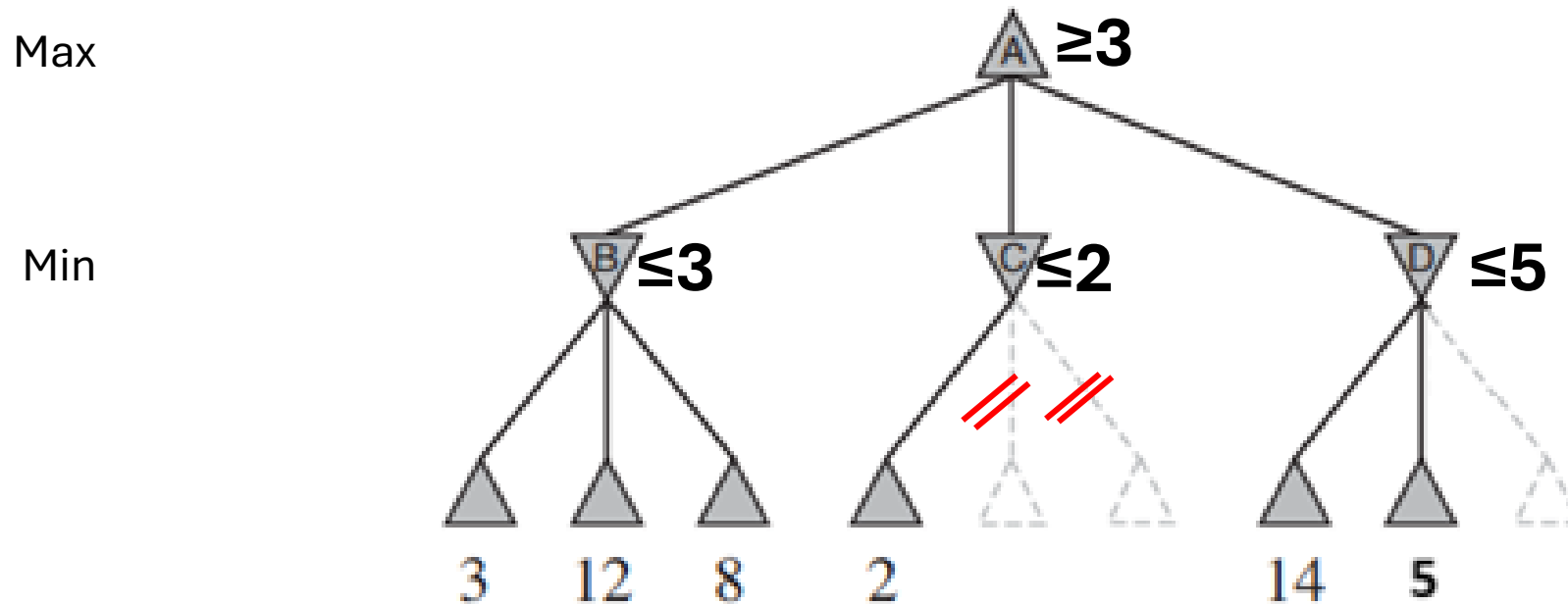
Alpha Beta Pruning



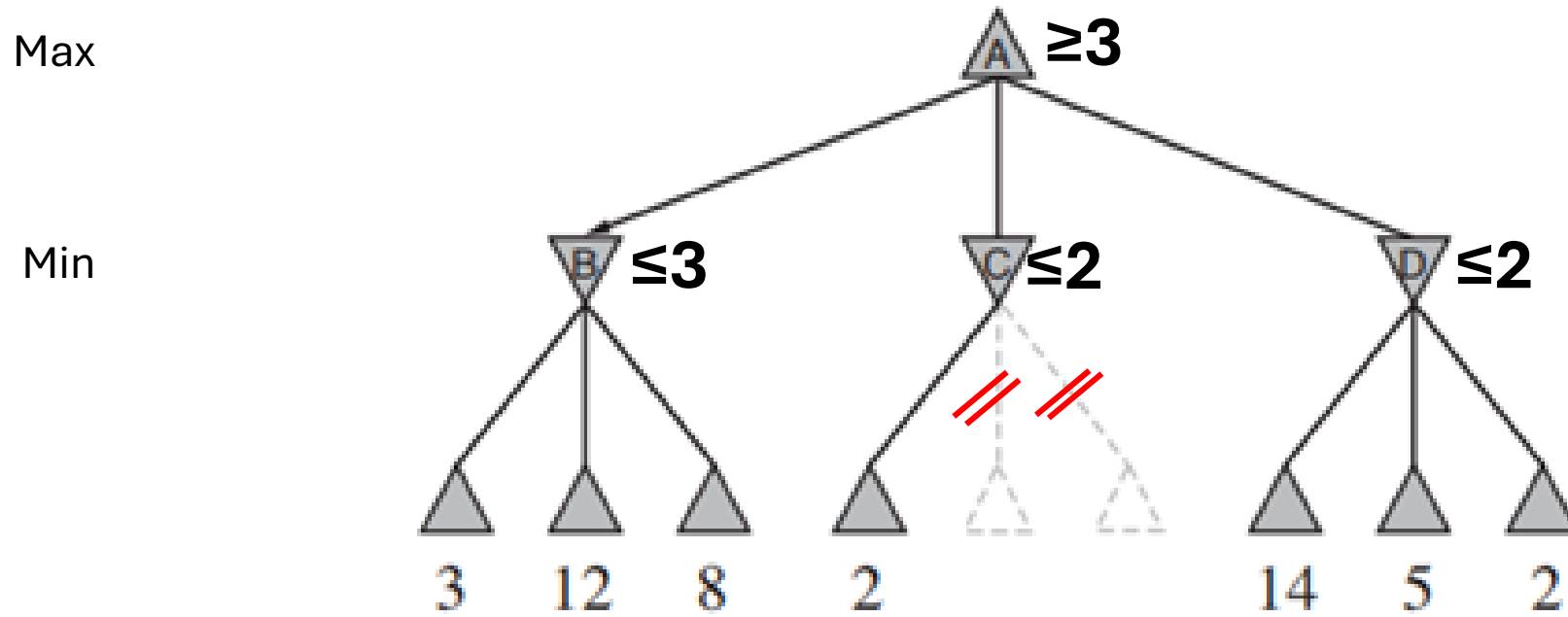
Alpha Beta Pruning



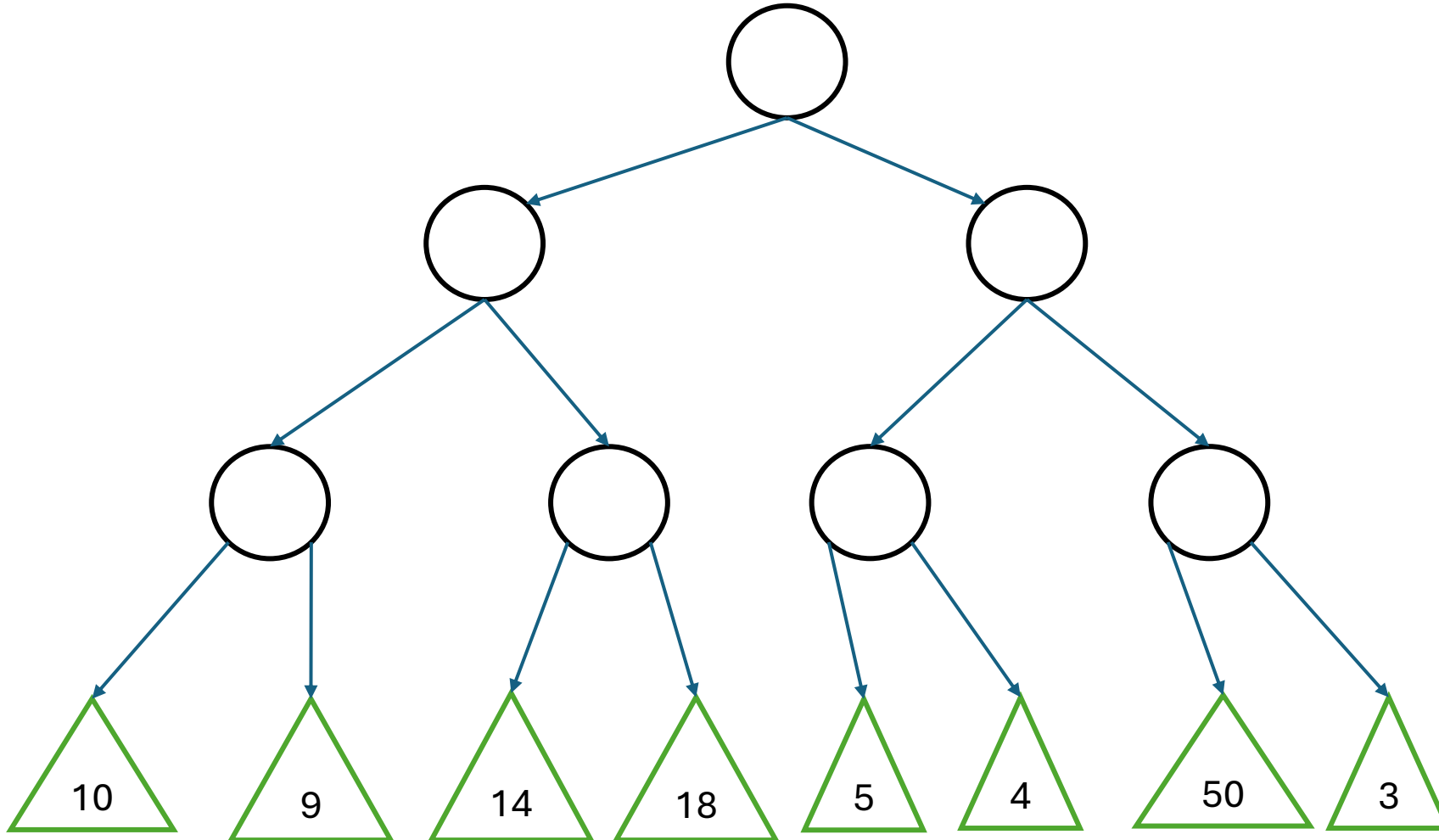
Alpha Beta Pruning



Alpha Beta Pruning



Alpha beta Pruning



References

- **CH 3- Solving Problems by Searching (Section 3.1-3.4.5)**